

Nathan L. Butler

(+1) 319-330-6155 | nathanbutler.nlb@gmail.com | [linkedin.com/in/nlbutler](https://www.linkedin.com/in/nlbutler) | natbut.github.io

ABOUT ME

Motivated roboticist with a background in artificial intelligence and mechanical design, seeking to develop scalable, generalizable autonomy solutions for real-world environments. Experienced in machine learning, robot autonomy, and rapid prototyping for field-deployable platforms.

EDUCATION

Oregon State University Sep. 2023 - Sep. 2025

M.S. in Robotics; minor in Artificial Intelligence

Iowa State University

Aug. 2018 - May 2023

B.S. in Mechanical Engineering; minors in Computer Science and Cyber-Physical Systems

GPA: 3.94/4.00; Awarded Outstanding Senior in Mechanical Engineering

RELEVANT COURSEWORK

Intro to Robotics I (Foundations) & II (Robotics Stack); Deep Learning; Sequential Decision Making; Multiagent Systems; Learning-Based Control; Intelligent Agents and Decision Making; Kinematics, Dynamics, and Controls; Machine Learning for Cyber-Physical Systems; Principles of Artificial Intelligence; Digital Logic

EXPERIENCE

Graduate Research Assistant, [Robotic Decision Making Lab](#), Oregon State Univ. Sep. 2023 - Sep. 2025

Machine Learning & Robotics Research

- Developed a hybrid decentralized planning algorithm enabling multi-robot teams to integrate local and global information for improved coordination in dynamic environments, with research contributions published at ICRA 2025.
- Implemented and trained a transformer-based model for multi-robot behavioral coordination in communication-constrained environments, providing scalable action guidance for robotic teams.
- Collaborated with multi-university teams on a grant-funded project to develop coordination algorithms for underwater robots, leveraging reinforcement learning for autonomous decision-making.
- Designed and integrated a modular hardware/software package that supports onboard autonomy (ROS 2 + ArduPilot) in autonomous surface vehicles, with applications in marine environments.

Undergraduate Research Assistant, [ABE Automation and Robotics Lab](#), Iowa State Univ. Jan. 2022 - Jul. 2023

Mechanical Design & Robotics Hardware

- Designed, sourced, and assembled custom electrical and mechanical components for heat dispersion and weatherproofing for outdoor stereo cameras, integrating 12 units into field robots for improved environmental robustness.
- Led the mechanical, electrical, and software integration of a robotic arm into a mobile field robot system, enabling dexterous crop data sampling in agricultural environments.
- Developed and implemented a modular design to decouple robotic sampling system from a mobile base, reducing recharging downtime and enhancing system operational efficiency.
- Provided timely mechanical, electrical, and software support for research team to successfully accomplish robotic field trials within constrained agricultural time windows.

Mechanical Engineer Co-Op, Seed Technology & Innovation Team, [Corteva Agriscience](#) May 2021 - Dec. 2021

Mechanical & Manufacturing Engineering

- Collaborated with engineering team and seed scientists to design, implement, and test automation solutions for internal research facilities.
- Designed, sourced, and assembled mechanical solutions for grasping and manipulation tasks for large, automated processing assemblies, integrated and evaluated prototypes CAD and in physical manufacturing settings.
- Produced and documented detailed mechanical component drawings, assembly design schematics, BOMs, and ECNs for internal record keeping, adhering to company and GD&T standards.
- Upgraded capabilities of computer vision-based product verification station, resulting in an 8x increase in throughput.

System Modeling & Control

- Modeled physical responses of electrical hardware components in Simulink for NASA's [Electrical Modeling and Thermal Analysis Toolbox](#) for use in physics-based digital twins of electric aircraft propulsion systems
- Modeled physical responses of electrical hardware components in Simulink for NASA's Electrical Modeling and Thermal Analysis Toolbox, supporting the development of digital twin models for electric aircraft propulsion systems.
- Assembled control models and tutorial documentation to assist NASA engineers and external researchers in using the Electrical Modeling and Thermal Analysis Toolbox for simulation and analysis tasks.

Systems Director, Cardinal Space Mining Club @ ISU

Aug. 2021 - May 2023

End-to-End Robotics Engineering

- Spearheaded team systems engineering activities including customer requirements development, interface tracking, tradeoff studies, and verification/validation testing, resulting in first-place finish at national Robotic Mining Competition.
- Managed compressed schedule and tight deadlines using data-driven problem-solving approaches to guide engineering subteams in decisions necessary for meeting engineering requirements.
- Coordinated team material & component procurement with local and national suppliers to support system fabrication and assembly within budget constraints.
- Led mechatronic design and assembly of material storage and offload subsystems for two competition robots, increasing onboard storage capacity and unload rate.
- Developed state machine for autonomous robot mining operations, rigorously tested and validated through top-performing mining operations in national competition.

Additional Projects:

- [MERL for Constrained Coordination](#): Developed a Multiagent Evolutionary Reinforcement Learning (RL) framework to train agents in tightly coupled tasks with sparse rewards, enhancing coordination for multi-robot systems.
- [Bravo MPC](#): Implemented Model Predictive Control (MPC) strategies for robotic arm task scheduling, optimizing task efficiency and real-time performance in robotic operations.
- [DQN for Task Scheduling](#): Created a Deep Q-Network (DQN) for efficient multi-robot task scheduling, optimizing resource allocation and task execution across robotic systems.
- **Robot Moisture Sensor**: Created hardware and software (Arduino + ROS 1) package for soil moisture probe, enhancing robotic field data collection opportunities.
- [Crop Row Robot Steering](#): Applied Autoencoder (AE) + Convolutional Neural Network (CNN) for visual-based steering commands, enhancing autonomous navigation within crop rows.

SKILLS & TOOLS

Software: Python (+PyTorch), ROS1/ROS2, C, Java, Linux, SolidWorks, MATLAB/Simulink, VSCode GitHub, Docker

Mechanical: System & Component Design, Engineering Documentation (Part/Assembly Drawings, BOMs, GD&T), DFA/DFM, Rapid Prototyping, Additive Manufacturing

Algorithms & Control: Learning-Based Methods (DL, RL), Multiagent Systems, Evolutionary Algorithms, Path Planning, MPC, PID, IK

Soft Skills: Robotic Frameworks, Mechanical Design, Systems Engineering, Project Management, Research & Analysis, Technical Documentation, Technical Communication